

**Solution:**

(a)

$$\begin{array}{rclcl}
 z & = & & x_1 & + & x_2, \\
 \hline
 w_1 & = & 27 & - & 3x_1 & - & 2x_2, \\
 w_2 & = & 8 & & & - & x_2, \\
 w_3 & = & 35 & - & x_1 & - & 5x_2, \\
 w_4 & = & 17 & - & 2x_1 & - & x_2,
 \end{array}$$

The initial solution is  $(x_1, x_2, w_1, w_2, w_3, w_4) = (0, 0, 27, 8, 35, 17)$ .

*Iteration 1.* Choose  $x_1$ , compute ratios:

$$\frac{27}{3} = 9, \quad \frac{35}{1} = 35, \quad \frac{17}{2} = 8.5,$$

Minimum is 8.5, so  $w_4$  leaves.

$$w_4 = 17 - 2x_1 - x_2 \implies x_1 = \frac{17}{2} - \frac{1}{2}w_4 - \frac{1}{2}x_2.$$

Substitute into all other equations:

$$\begin{array}{rclcl}
 z & = & \frac{17}{2} & - & \frac{1}{2}w_4 & + & \frac{1}{2}x_2, \\
 \hline
 w_1 & = & \frac{3}{2} & + & \frac{3}{2}w_4 & - & \frac{1}{2}x_2, \\
 w_2 & = & 8 & & & - & x_2, \\
 w_3 & = & \frac{53}{2} & + & \frac{1}{2}w_4 & - & \frac{9}{2}x_2, \\
 x_1 & = & \frac{17}{2} & - & \frac{1}{2}w_4 & - & \frac{1}{2}x_2.
 \end{array}$$

New solution is  $(x_1, x_2, w_1, w_2, w_3, w_4) = (\frac{17}{2}, 0, \frac{3}{2}, 8, \frac{53}{2}, \frac{17}{2})$ .

*Iteration 2.* Choose  $x_2$ , compute ratios:

$$\frac{3/2}{1/2} = 3, \quad \frac{8}{1} = 8, \quad \frac{53/2}{9/2} = \frac{53}{9}, \quad \frac{17/2}{1/2} = 17,$$

Minimum is 3, so  $w_1$  leaves.

$$w_1 = \frac{3}{2} + \frac{3}{2}w_4 - \frac{1}{2}x_2 \implies x_2 = 3 + 3w_4 - 2w_1.$$

Substitute into all other equations:

$$\begin{array}{rclcl}
 z & = & 10 & + & w_4 & - & w_1, \\
 \hline
 x_2 & = & 3 & + & 3w_4 & - & 2w_1, \\
 w_2 & = & 5 & - & 3w_4 & + & 2w_1, \\
 w_3 & = & 13 & - & 13w_4 & + & 9w_1, \\
 x_1 & = & 7 & - & 2w_4 & + & w_1.
 \end{array}$$

New solution is  $(x_1, x_2, w_1, w_2, w_3, w_4) = (7, 3, 0, 5, 13, 0)$ .

*Iteration 3.* Choose  $w_4$ , compute ratios:

$$\frac{5}{3}, \quad \frac{13}{13} = 1, \quad \frac{7}{2},$$

Minimum is 1, so  $w_3$  leaves.

$$w_3 = 13 - 13w_4 + 9w_1 \implies w_4 = 1 - \frac{1}{13}w_3 + \frac{9}{13}w_1$$

Substitute into all other equations:

$$\begin{array}{rclcl} z & = & 11 & - & \frac{1}{13}w_3 & - & \frac{4}{13}w_1, \\ x_2 & = & 6 & + & \frac{3}{13}w_3 & + & \frac{1}{13}w_1, \\ w_2 & = & 2 & + & \frac{3}{13}w_3 & - & \frac{1}{13}w_1, \\ w_4 & = & 1 & - & \frac{1}{13}w_3 & + & \frac{9}{13}w_1, \\ x_1 & = & 5 & + & \frac{2}{13}w_4 & - & \frac{5}{13}w_1. \end{array}$$

Thus, we obtain the optimal solution  $(x_1^*, x_2^*) = (5, 6)$ ,  $z^* = 11$ .

(b) Introduce dual variables  $y_1, y_2, y_3, y_4 \geq 0$  for the four constraints. The dual is

$$\begin{array}{ll} \min & 27y_1 + 8y_2 + 35y_3 + 17y_4, \\ \text{s.t.} & 3y_1 + 0y_2 + y_3 + 2y_4 \geq 1, \\ & 2y_1 + 1y_2 + 5y_3 + 1y_4 \geq 1, \\ & y_1, y_2, y_3, y_4 \geq 0. \end{array}$$

From part (a), we have optimal solution  $(x_1^*, x_2^*) = (5, 6)$  and  $z^* = 11$ . Thus,  $27y_1 + 8y_2 + 35y_3 + 17y_4 \geq x_1^* + x_2^* = 11$ . Using the same method in part (a), we obtain the optimal dual solution  $(y_1^*, y_2^*, y_3^*, y_4^*) = (\frac{28}{91}, 0, \frac{7}{91}, 0)$ .